

He Min

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EDUCATIONAL BACKGROUND

Nanyang Technological University

Master of Science in Signal Processing and Machine Learning

Chongqing University of Technology

Bachelor of Science in Computer Science and Technology (Honours)

• GPA: 3.96 / 5.0 (Top 1%)

Singapore

2026.01-2027.07

Chongqing, China

2021.09-2025.06

INTERNSHIP EXPERIENCE

Lenovo (Shanghai) Information Technology Co., Ltd. (LR Robotics Algorithm)

2026.05-2026.08

Research Intern in Embodied AI for Robotics

Project Overview: Leveraging the Lenovo Daystar robot as the platform, a collaborative intelligent agent system has been constructed, comprising the task-planning "mastermind" **NavAgent** and the dual-system navigator **InternVLA-N1-DualVLN**. NavAgent utilizes the **LangGraph** state machine and **MCP** to orchestrate the **ROS2** toolkit, handling global task parsing, **memory** storage, telemetry **self-checks**, and anomaly diagnosis. **DualVLN** manages specific motion generation: **System 2** (2 Hz; **fine-tuned QwenVL-2.5**) predicts 2D pixel sub-goals, while **System 1** (30 Hz; **NextDiT**) outputs high-frequency physical trajectories. **SR** increased by **10.1%**, **SPL** increased by **6.7%**, **NE** decreased by **17.6%**, and **StR** decreased by **32.5%**.

PROJECT EXPERIENCE

Fine-tuning and Closed-loop Evaluation of SmolVLA Robot Fruit-Grasping Strategies Based on Multimodal Synthetic Data[\[github\]](#)

2026.07- Present

Project Overview: This project established an end-to-end development pipeline—spanning multi-objective synthetic data generation, **VLA** policy fine-tuning, and closed-loop evaluation—on **AMD GPUs (ROCm)**. Leveraging the **Genesis** IK expert controller, the team autonomously generated and aggregated a standard **LeRobot multimodal** dataset comprising **150 episodes** (over **30,000 frames**) featuring three categories of geometric fruit. Building on this, **SmolVLA** underwent imitation learning fine-tuning, achieving a **76% success rate** in end-to-end closed-loop placement tasks within **environments characterized by cross-domain visual and physical disturbances** (an 8% improvement over models without domain randomization).

OV-LidarMap: Open-Set Instance-Level 3D Semantic Mapping with Rotating LiDAR and Panoramic Vision

Co-first Author, **IEEE RA-L (SCI Q2 TOP, Under Review)** [\[arXiv\]](#) Supervisor: Prof. Lihua Xie 2026.01-2026.03

Project Overview: To address the lack of spatial reasoning capabilities in LLMs and the issue of context explosion within large-scale 3D environments, we developed an omnidirectional perception-driven embodied **AI agent** framework using **Python**, **PyTorch**, and **ROS 2**. The front-end integrates rotating LiDAR and **panoramic vision** to construct a **five-layer dynamic scene graph**, utilizing persistent homology-based adaptive room segmentation and VLM-based hybrid pruning. At the decision-making level, we pioneered the use of **3D 8-neighbor connectivity** and **multi-resolution spatial attention prompting** to achieve **zero-shot long-range navigation**; this resulted in a **93.18% success rate** (with a **22.73%** improvement in viewpoint-dependent accuracy) and a **69.98% reduction in token costs**.

PUBLICATION

He, M. (2025). A Review of Data Fusion and Deep Learning Models for Multimodal Sentiment Analysis. In Proceedings of E-commerce and Artificial Intelligence (ECAI 2024). [\[SciTePress\]](#)

AWARDS & HONORS

National Scholarship (2024); First Prize, National College Student Mathematics Competition (2023); Third Prize, National Finals of the Team Programming Ladder Tournament (2023)

TECHNOLOGY STACK

• **Embodied Intelligence & Large Model Agent:** Proficient in **VLA (Vision-Language-Action)** end-to-end policy fine-tuning and imitation learning; skilled in **LangGraph** state machine design and encapsulation of **MCP** Protocol tool ecosystems; familiar with **CoT** reasoning chains and large model adaptive resolution / prompt compression technologies.

• **Robotic Systems, Perception and Simulation:** Proficient in **ROS 2** communication mechanisms; familiar with physics simulation engines such as **Genesis** and the **LeRobot** embodied dataset standard; experienced in **SAM2** object segmentation, **Open3D** point cloud processing and **KDTree** search; with hands-on experience in constructing 3D **dynamic scene graphs**.

• **Systems Engineering & Concurrent Programming:** Proficient in **Python** and **C++**; familiar with **Linux**, **Docker** containerization, and **Git** version control; experienced in encapsulating **low-level control logic** for **multi-threaded applications** and conducting **hardware-software integration and debugging**.